

Original citation:

Tkacz, Nathaniel. (2012) From open source to open government : a critique of open politics. *Ephemera: Theory and Politics in Organization*, Vol.12 (No.4). pp. 386-405. ISSN 1473-2866

Permanent WRAP url:

<http://wrap.warwick.ac.uk/53295>

Copyright and reuse:

The Warwick Research Archive Portal (WRAP) makes the work of researchers of the University of Warwick available open access under the following conditions.

This article is made available under the Creative Commons Attribution-NonCommercial-NoDerivs 3.0 Unported (CC BY-NC-ND 3.0) license and may be reused according to the conditions of the license. For more details see: <http://creativecommons.org/licenses/by-nc-nd/3.0/>

A note on versions:

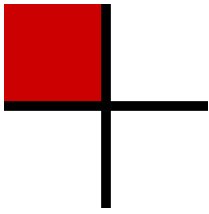
The version presented in WRAP is the published version, or, version of record, and may be cited as it appears here.

For more information, please contact the WRAP Team at: wrap@warwick.ac.uk

warwick**publications**wrap

highlight your research

<http://go.warwick.ac.uk/lib-publications>



From open source to open government: A critique of open politics

Nathaniel Tkacz

abstract

Notions of openness are increasingly visible in a great number of political developments, from activist groups, software projects, political writings and the institutions of government. And yet, there has been very little reflection on what openness means, how it functions, or how seemingly radically different groups can all claim it as their own. Openness, it seems, is beyond disagreement and beyond scrutiny. This article considers the recent proliferation of openness as a political concept. By tracing this (re)emergence of 'the open' through software cultures in the 1980s and more recently in network cultures, it shows how contemporary political openness functions in relation to a new set of concepts – collaboration, participation and transparency – but also identifies important continuities with previous writings on the open, most notably in the work of Karl Popper and his intellectual ally Friedrich Hayek. By revisiting these prior works in relation to this second coming of the open, the article suggests that there is a critical flaw in how openness functions in relation to politics, beginning with the question: How is that new movements championing openness have emerged within a supposedly already-open society?

Introduction

Most think about these issues of free software, or open source software, as if they were simply questions about the efficiency of coding. Most think about them as if the only issue that this code might raise is whether it is faster, or more robust, or more reliable than closed code. Most think that this is simply a question of efficiency. Most think this, and most are wrong ... I think the issues of open source and free software are fundamental in a free society. I think they are at the core of what we mean by an open society.

– Lawrence Lessig (2005: 260)

One approach to understanding the democracy of the multitude, then, is as an open-source society, that is, a society whose source code is revealed so that we all can work collaboratively to solve its bugs and create new, better social programs.

– Michael Hardt and Antonio Negri (2004: 340)

‘The open’ has become a master category of contemporary political thought. Such is the attraction, but also the ambiguity of openness, that it appears seemingly without tension, without need of clarification or qualification, in writers as diverse as the liberal legal scholar, Lawrence Lessig, and the post-Marxian duo Michael Hardt and Antonio Negri. Every political position worth its salt, it seems, must today pledge allegiance to this strange and relatively new political concept. The epigraphs above are indicative of a development that forms the basis of this paper: the re-emergence and re-politicisation of openness in relation to a set of developments specific to the realm of software. In the first epigraph, Lessig looks back, trying to connect open source and free software to an already existing notion of open politics, ‘the open society’. Hardt and Negri – who, it must be said, are a long way from home on this matter – look forward, trying to establish a connection between really existing practices and logic of open source software, and their yet to be realised ‘democracy of the multitude’. As does Lessig, I begin this paper by connecting back, by revisiting the father of open thought, Karl Popper. I trace what might be called the second coming of the open, through debates about open systems and open software in the 1980s and 90s, to the generalisation and proliferation of openness in network cultures – evidenced by such notions as open access, open education and open communities – and finally, to the re-emergence of the open in institutional politics and related writings.

My purpose is not to pin down the meaning of openness, nor to moralise upon this notion, but rather to trace its proliferation and consider how it functions in contemporary cultures, the writings of Popper, and in relation to competing and supporting concepts. Through a reconsideration of Popper, I finish by outlining some concerns for contemporary proponents of open politics – a task that I consider crucial as the open is increasingly used to ‘look forward’.

The open society

Karl Popper was not the first to write about the concept of openness, nor even of the open society (Bergson, Audra et al., 1935; Bertalanffy, 1960). However, it wasn’t until Popper wrote *The open society and its enemies* (1962), while in exile in New Zealand during the Second World War, that the political notion of the open gained mass appeal. In two volumes, Popper rewrites the history of political philosophy, and also lived political conflict, around the concept of openness. He locates the origins of his version of ‘the open society’ in the ‘breakdown of Greek tribalism’ (Popper, 1962: 183), culminating in the Peloponnesian War (circa 431-404 BC) between the Delian League, headed by Athens, and the Peloponnesian League, led by Sparta. Interwoven with this history is a detailed critique of Plato, whose ‘closed’ political philosophy, Popper argues, is strongly marked by these events. Plato is depicted as a brilliant but misguided thinker whose experiences of the war (and especially the execution of his mentor Socrates) lead him to build a totalitarian and reactionary political philosophy. This philosophy, Popper writes, is built on the principle that virtually all change is bad, and society, which is always ‘in flux’, is therefore in a state of deterioration. In opposition to this state of flux, Plato posits an original ideal form of

society existing in ancient history, highly stable and resistant to change from which the current imperfect society is derived. This original state equates to the *theory of forms or ideas* that underpins Plato's philosophical thought: the original tribal society is the ideal, whereas the actually existing society, with all its problems, is the inferior and degraded version of this form. In the battle between Athens and Sparta, the older, 'tribal' Spartan social structure is considered more desirable as it is closer to the ideal form, while the Athenian democracy represents radical change and, therefore, degeneration. It is around this notion of negative change and the ideal ancient Greek tribal form that Plato writes *The Republic* (1974). *The Republic* describes a society where all change is arrested. The social is organised around three classes – rulers, auxiliaries and producers – all with highly specific and unchanging roles. The whole social edifice – education, law, reproductive norms and so on – is designed to maintain this strict demarcation and rigid order. There is no 'cross-breeding' between the classes and social interaction between them is avoided.

Philosophy, conceived as the perception of ideal forms, emerges in Plato's thought as the bridging device from the status quo to this ideal state. As the famous 'simile of the cave' passage reminds us, Plato posits the philosopher as the only actor able to see true knowledge – the light of the sun as opposed to the darkness and shadow puppets on the cave wall – and thus, as the only individual qualified to determine how a society should be organised. Such enlightenment also distances the philosopher from the desires and indulgences of everyday life and thus makes them even more suitable rulers of society – so-called Philosopher Kings.

Popper critiques Plato on multiple grounds but the overall argument can be summarised as follows: Plato claims to possess a kind of true knowledge, the knowledge of forms. This knowledge provides the general laws of history and at the same time positions the philosopher as the only person able to steer society in the right direction (because of the knowledge they possess about how things should be). All decision-making capacity is removed from everyone except the philosopher, who decides in the most disinterested fashion what is right for all. That is, armed with the knowledge of history, with its ineluctable laws, the philosopher is almost compelled to become a *social engineer*. Deprived of any capacity to choose due to the reification of all roles and duties, coupled with the subjugation of non-philosophical knowledge – the mere 'knowledge of shadows' – individuals in Plato's *Republic* are effectively denied agency.

Popper's critique of closed thought and politics extends well beyond the writings of Plato. Any political philosophy based on unchallengeable truths – such as the discovery of the *laws of history* – that provide definite and rigid future programmes, and where individual will is always subordinated to these larger truths, is described in the language of the *closed society*. For Popper, the three most important philosophers in this tradition after Plato are Aristotle, Hegel and Marx. Aristotle is largely dismissed as Plato's mouthpiece, with the exception that he puts a positive spin on Plato's theory of forms: rather than constantly degrading, the state is positioned as heading towards an ultimate end, towards perfection. Aristotle is important for Popper, however, because his biologically influenced teleological

thought is taken up by Hegel, which in turn informs German nationalism through the notion of the destiny of one race (the most perfect) to rule all others, as well as Marx's laws of class struggle and the destiny of the proletariat. Thus, Plato is significant not only as the first closed thinker or 'enemy of the open society', and not just because he influenced these key historical figures, but because it is his political philosophy that informed the two major competing political programs during the second world war: fascism and communism. In Popper's time, therefore, fascism and communism are the modern manifestations of the closed society, while capitalism and the democratic institutions affiliated with it represent the open society.

The summation of Popper's thought is a re-articulation of existing political concepts (democracy, fascism, communism), of the writings of key historical figures of political philosophy (Plato, Aristotle, Hegel, Marx and others) and of lived conflict (the Peloponnesian War, WW2) around the new master categories of open and closed. In this new politics of the open/closed, the fate of a nation and its people, or alternatively the class inequalities produced by capitalism, are no longer the primary concern. The question is no longer about identity, race or class, but whether or not a social programme, that is, a set of knowledges and related practices, is able to change. Social programmes based on unchallengeable truths – the so-called laws of history or of destiny – emerge as the fundamental enemy, and what might be considered radically different political programmes in a different frame of analysis – communism and fascism – are made equivalent. The positive side of this political equation, the open society, is one where totalising knowledge is necessarily impossible. Openness is necessary because nobody can know for certain what the best course for society might be from the outset, and at the same time it is assumed that openness provides the best possible conditions for producing knowledge and, therefore, making better decisions.

I return to Popper and the open society below, but first I want to map the re-emergence and re-articulation of openness, beginning in software cultures, through to network cultures and more traditional political institutions. I want to demonstrate the significance of openness by gesturing toward its proliferation and showing how it is increasingly held up as the highest political ideal.

From systems to source, or, how we became open (again)

By the 1980s, the USA was under the sway of neo-liberalism. The organisational philosophy of 'competition' had seemingly defeated the socialist desire for 'centralised planning' in the socioeconomic ideology wars. The revised philosophical argument championing liberalism was put forward by Friedrich Hayek (1944) and resonates strongly with Popper's notion of the open society. Hayek argues that the knowledge of how a society should be organised and which direction it should take is beyond any one individual or group and can never be known with certainty. Because of this, any attempt at centralised planning (i.e. socialism, communism, fascism) which is founded on exactly the assumption

that what is best for all society is directly knowable, is likely to produce bad decisions that only satisfy a small group. For Hayek, giving one group the ability to make decisions for the whole results in the overall reduction of liberty and the advent of totalitarianism. Instead, Hayek suggests, once society reaches a certain complexity only a decentred mode of organisation, where competing ideas and practices can interact and adjust in relation to change, can ensure liberty:

It is only as the factors which have to be taken into account become so numerous that it is impossible to gain a synoptic view of them, that decentralisation becomes imperative. ... decentralisation has become necessary because nobody can consciously balance all the considerations bearing on the decisions of so many individuals, the co-ordination can clearly not be affected by 'conscious control', but only by arrangements which convey to each agent the information he must possess in order effectively to adjust his decisions to those of others. (Hayek, 1944: 51)

The precise form that decentralisation takes is competitive markets. Such markets theoretically enable many individuals to shape society through the sale and purchase of commodities and, thus, with no 'conscious control'. Freedom is therefore intimately tied to economic freedom – to the freedom to sell commodities, including human labour, in a market. But the argument for economic freedom derives from a more fundamental critique of knowledge and centralisation. Thus, the critique of totalitarian knowledge put forward by Popper, and shared by Hayek, is translated into government and economic policy to justify competitive, market-based forms of organising society.

With these larger changes in the theory and practice of governance taking place in the background, important new contests over openness arose in computer cultures, specifically around the notions of open systems and, soon after, software. These contests were seemingly far removed from Hayek-inspired neo-liberal agendas but, as we shall see, arguments made by Popper and Hayek at the level of philosophy and economics are isomorphic with the ones that played out in computer cultures. In regard to systems, Christopher Kelty has covered how debates about openness played out around the UNIX operating system as well as the TCP/IP protocols. He describes these debates as at once technical and moral, 'including the demand for structures of fair and open competition, antimonopoly and open markets, and open standards processes...' (Kelty, 2008: 144). In the open systems debates, the battle for openness is not against closed forms of knowledge, à la Popper, but against proprietary standards – what might be described as closed infrastructures. I will not recount in detail this battle for open systems (this has already been done very well by Kelty). Instead, I will focus on one story that developed throughout this period: the birth of Free Software and the challenge of Open Source. I focus on this story because it surpasses notions of openness in open systems and captures both the lived experience and the contested distributions of agency in software cultures. It reveals how competing mutations of liberalism were aligned with new legalities and modes of production and, most importantly, how all these developments would redefine and re-energise political openness.

In 1980, a group of adventurous programmers at MIT, including a young Richard Stallman, were confronted with a problem: the AI Lab they were working in had received a new Xerox 9700 Laser Printer. As the printer station was located on a different floor to the majority of people who used it, the young Stallman had written a small programme for the previous Xerographic Printer that electronically notified a user when their print job was finished and also alerted all logged in users when the printer was jammed. This required some minor modifications of the Xerographic Printer's code. When the new Xerox machine arrived, Stallman intended to make similar programme modifications. But curiously, this new machine, which was offered to the lab as a 'gift' from Xerox, did not arrive accompanied with a document containing the printer's (human readable) source code. Without the source code, no modifications could be made to the Xerox. Stallman decided to track down the original programmer from Xerox to ask for the source code personally. On confronting the programmer, he was told that he could not have a copy of the source code and, moreover, that the programmer had signed a Non Disclosure Agreement (NDA), which at the time was a complete novelty in the field of software. It was after this encounter with Xerox, the story goes, that Stallman famously declared, 'All software should be free'. Not (only) in the sense of free to use or free to distribute, but in that greater sense of free to change, modify, rewrite, adapt... – in short, a freedom to reorganise and modify the algorithms that instruct the machines that populate our worlds. This story circulates as the mythic origin of free software and establishes Stallman as its guru and prophet.

After growing increasingly disillusioned with the effects of commodification as a mechanism of control taking place both in his immediate environment and the wider software community, Stallman left the lab at MIT. His plan was to create an entire operating system (OS) that would not be subject to what he perceived as the *closure* of proprietary software. In 1983 he announced plans to create the GNU OS as part of a new Free Software Movement (FSM). The GNU OS was to be written from scratch using non-proprietary code. In 1985, he set up a non-profit corporation called the Free Software Foundation (FSF) to formally oversee and represent the movement. The most significant development during this period, however, was the creation of several unique copyright licences designed to keep the outputs of the FSM 'open'. Although, initially, specific licenses were written for each new piece of software, in 1989 Stallman developed the GNU General Public License (GNU GPL) as a broadly applicable software license. These licenses are generally described as using the mechanism of copyright against itself, in that rather than restricting distribution through the creation of scarcity they use copyright to ensure an application or text can be accessed, made visible, dissected and modified or 'remixed' (Lessig, 2008). The GNU licences were not the only permission-based (as opposed to restriction-based) licenses, but the GNU GPL in particular was certainly the most progressive of its type; not only was any piece of software created under it accessible and modifiable, but the license states that any derivative of an earlier text/program must also adopt the same license. This was the legal mechanism that supported Stallman's desire to keep the outputs of his FSM 'free' and the movement as a whole growing (because of the so-called viral nature of the license). The FSF would not only oversee the movement:

Stallman also suggested that any product making use of the license be signed over to the foundation, which would police any violations and take appropriate legal action.

While Stallman proclaims that code is necessarily political, other programmers have attempted to uncouple this pairing (see Berry, 2008: 147-187; Williams, 2002: 136). Indeed, Gabriella Coleman (2004: 3) has argued that the refusal to acknowledge their actions as political is one of the key characteristics of many software cultures. In 1998, a group of high profile programmers started the Open Source Initiative (OSI). The most vocal member of this group, Eric Raymond, viewed Stallman as a controlling idealist who focused too much on politics at the expense of technical excellence and efficiency. This strategic reframing of the question of code in terms of excellence and efficiency was designed to make free software business friendly. The term Open Source was chosen to sidestep the connotations surrounding Stallman's rhetoric of 'free', which seemed less than appealing to profit-seeking enterprises, especially when attached to a product. 'Open Source' is well chosen as it foregrounds the technical dimension of these software practices – 'this movement is about source code' – and conveniently sidesteps Stallman's political concerns. In order to achieve this distance from the FSM, the OSI had to generate their own licenses that effectively reversed the 'viral' nature of the GPL. The challenge for these licenses – such as the Mozilla Public License – was to 'balance' the requirement for companies to commodify software outputs with the increased potential for productivity, made possible by involving outsiders and harnessing their 'contributions'.

The OSI also had its own gurus in Linus Torvalds and Eric Raymond. In 1997 Raymond published his first iteration of *The cathedral and the bazaar*, a hugely influential musing on the production method he observed in the Linux operating system – the project headed by Torvalds. The terms 'cathedral' and 'bazaar' are used to represent competing production methods. Of the cathedral method, Raymond writes:

I had been preaching the Unix gospel of small tools, rapid prototyping and evolutionary programming for years. But I also believed there was a certain critical complexity above which a more centralized, a priori approach was required. I believed that the most important software (operating systems and really large tools like the Emacs programming editor) needed to be built like cathedrals, carefully crafted by individual wizards or small bands of mages working in splendid isolation... (Raymond, 2000: 3)

The bazaar mode of production, found in Linux, emerges as the improbable yet superior other:

Linus Torvalds's style of development – release early and often, delegate everything you can, be open to the point of promiscuity – came as a surprise. No quiet, reverent cathedral-building here. Rather, the Linux community seemed to resemble a great babbling bazaar of differing agendas and approaches (aptly symbolized by the Linux archive sites, who'd take submissions from anyone) out of which a coherent and stable system could seemingly emerge only by a succession of miracles.

The fact that this bazaar style seemed to work, and work well, came as a distinct shock. As I learned my way around, I worked hard not just at individual projects, but also at trying to understand why the Linux world not only didn't fly apart in confusion but seemed to go from strength to strength at a speed barely imaginable to cathedral-builders. (*ibid.*)

From open source to open government: A critique of open politics Nathaniel Tkacz

Out of the FSM and the OSI emerge two competing mutations of liberalism. With Stallman lies the recognition that the creation of markets via the commodification of software actually reduces the capacities (or liberties) of individuals who use and modify it. The argument for open markets that played out in the open systems debates is extended to software itself. It is a liberal argument that fundamentally challenges the pre-existing liberal coupling of freedom and property. Openness is primarily understood as a techno-legal quality, whose opposite, as Kelty reminds us, ‘is not closed, but “proprietary”’ (2008: 143). With Raymond, on the other hand, the emphasis is not on commodification, but the organisation of production. Hayek’s argument about the ideal organisation of society, as described above, strongly parallels those put forward by Raymond, but at the level of individual contributions to specific software projects. The cathedral parallels the ‘centralised planning’ critiqued by Hayek, while the bazaar emerges as a new liberal utopia: radically open to competing ‘agendas and ideas’; progress ‘at a speed barely imaginable’; and the miraculous emergence of a ‘coherent and stable system’. The history of the OSI, and the writings of Raymond in particular, demonstrate how contemporary political openness came to be articulated with a specific method of software development. The open would come to be articulated alongside notions of participation, transparency and increased efficiency. While Stallman remained steadfast in his preference for the term ‘free’ to describe his movement and its outputs, it was the business-backed ‘open source’, and eventually just ‘open’, that captured the minds of people outside software culture.

The open takes flight

While software, such as GNU/Linux-based operating systems, the Apache server client and the Mozilla web browser, have made FLOSS highly visible within software communities for many years, it is the *translation* of these ideas into new domains that is most significant. The material covered in this section is by no means exhaustive. Rather, I focus on a series of examples that demonstrate certain specific qualities of how openness has been translated. I begin with projects that name themselves ‘open’ and thus explicitly interpret their activities in relation to openness. I then look at different projects that describe key facets of their activities in terms of openness, including activist groups that organise around openness, two ‘mainstream’ entities (Wikipedia and Google), and finally different political writings and government initiatives that make use of the open. To be sure, there are many significant differences between all the examples covered but my focus is on this very fact: that the same rhetoric is deployed by what are otherwise very different groups or organisations. It is the fact of diffusion that is most significant.

The most obvious translation of openness emerges from online projects or movements that explicitly name themselves as such. Within this category are broad movements or trends, such as Open Access, which is generally used to describe the making available of published content and especially scholarly, educational and scientific materials. One example of open-as-Open Access is the Open Humanities Press (2010). This initiative publishes academic monographs but also acts as a kind of branding or certification mechanism for a

series of online journals. Open Humanities Press has four stated principles that cover access, scholarship, diversity and transparency, and a series of related goals, the last of which is to ‘explore new forms of scholarly collaboration’ (2010). Other examples of Open Access include the Bentham Open project (2010); individual journals such as Open Medicine, which makes its content ‘freely available for others to read, download, copy, distribute, make derivative works (‘remix’) and use with attribution’ (2010) and whose stated mission includes the promotion of international ‘collaboration on health issues’ (2010); and finally, open access study material, such as those provided by participants in the Open Courseware Consortium (2010), of which MIT’s Open Courseware project (2010) is perhaps the best known.

Closely related to these open access initiatives are projects that include an open access component, but also emphasise a broader or perhaps more ‘procedural’ sense of openness. A good example here is The Open Knowledge Foundation. The Foundation ‘seeks a world in which open knowledge is ubiquitous and routine’ and sees openness as having ‘far-reaching societal benefits’ (2010). The Foundation states, for example, that politically ‘openness improves governance through increased transparency and engagement’; culturally, ‘openness means greater access, sharing and participation’; economically, ‘openness permits easier and more rapid reuse of material’; and ‘for science to effectively function, and for society to reap the full benefits from scientific endeavours, it is crucial that public scientific information be open’ (The Open Knowledge Foundation, 2010). Because of these perceived benefits, the Foundation supports and facilitates a range of projects, including the Open Data Commons, Open Shakespeare, Open Economics, Open Text Book, Open Milton, Open Knowledge Forums, Open Geodata and Open Environmental Data. I have listed only the projects with ‘open’ in the title, but there are numerous others (see The Open Knowledge Foundation, 2010).

The strongest expression of *translated openness*, however, is to be found in the Open Everything movement (2010). Open Everything has a wiki that details its events and its function. The welcome page of the wiki states:

Open Everything is a global conversation about the art, science and spirit of 'open'. It gathers people using openness to create and improve software, education, media, philanthropy, architecture, neighbourhoods, workplaces and the society we live in: everything. It's about thinking, doing and being open. (2010)

Further down the page are a series of statements about the open:

Open is changing the game. And, while Wikipedia and open source software offer great examples... we know that openness, collaboration and participation are spreading well beyond the realm of technology... Where open is headed is huge. Open Everything gathers people who are charting this trajectory. (2010)

Openness is conceived as a new mode of being, applicable to many areas of life and gathering significant momentum – ‘changing the game’ as it were. Once again, this ‘spirit of open’ is closely articulated with collaboration and participation. The Free and Open

Everything initiative is also associated with The Foundation for P2P Alternatives, which functions ‘as a clearing house for open/free, participatory/p2p and commons oriented initiatives’ (P2P Foundation Contributors, 2010). The P2P Foundation has its own Open Everything directory, including a detailed mind map titled “Everything Open and Free” (see Figure 1), which attempts to comprehensively map and classify the dimensions of openness.

Linked from the central “Everything Open and Free” hub is an array of different nodes, each of which covers a different dimension of openness, including: Aspects of Openness, Enablers of Openness, Infrastructures of Openness, Practices of Openness, Domains of Openness, Products of Openness, Open Movements and Open Consciousness. The Open Everything project, together with the Everything Open and Free mind map, represents an emerging desire to radically transform society around the concept of openness. Translated from the world of software (but not reducible to it), openness must therefore be understood as a powerful new form of political desire in network cultures.

This new stated commitment to the open is not limited to explicitly activist and marginal network cultures. Two radically different but equally ‘mainstream’ organisations, Wikipedia and Google, also understand their operation in terms of openness. As the ‘free encyclopaedia that anyone can edit’, Wikipedia is commonly held up as the most successful example of translated openness. On the “About” page of the English Wikipedia, the encyclopaedia is described as ‘open to a large contributor base’; ‘written by open and transparent consensus’; and the various effects of its ‘radical openness’ are considered (Wikipedia Contributors, 2010). Moreover, the project is built on wiki software, which allows for easy and immediate page creation and modification and is licensed under permissive, commons-based licenses (Creative Commons Attribution – Sharealike 3.0 License and the GNU Free Documentation License). Interestingly, while Wikipedia is more often celebrated as the open ideal in terms of contribution, governance, technology and licensing, the rhetoric of openness is stronger in Google’s case. For example, on Google’s *Public Policy Blog*, Senior Vice President of Product Management, Jonathan Rosenberg, published a post titled “The meaning of open”. He writes: ‘In an open system, a competitive advantage doesn’t derive from locking in customers, but rather from understanding the fast-moving system better than anyone else and using that knowledge to generate better, more innovative products’ (2009). The sentiments expressed in the post are very similar to the ones offered by Raymond and the Open Source Initiative, where openness is figured as an innovative and competitive production method perfectly compatible with a new form of capitalist accumulation. Rosenberg goes on to define openness in terms of hardware and software, information, transparency, and control. The key passage, however, comes toward the end of the post, where Rosenberg is explicit about what he sees is at stake in the battle for openness:

Open will win. It will win on the Internet and will then cascade across many walks of life: The future of government is transparency. The future of commerce is information symmetry. The future of culture is freedom. The future of science and medicine is collaboration. The future of entertainment is participation. Each of these futures depends on an open Internet. (2009)

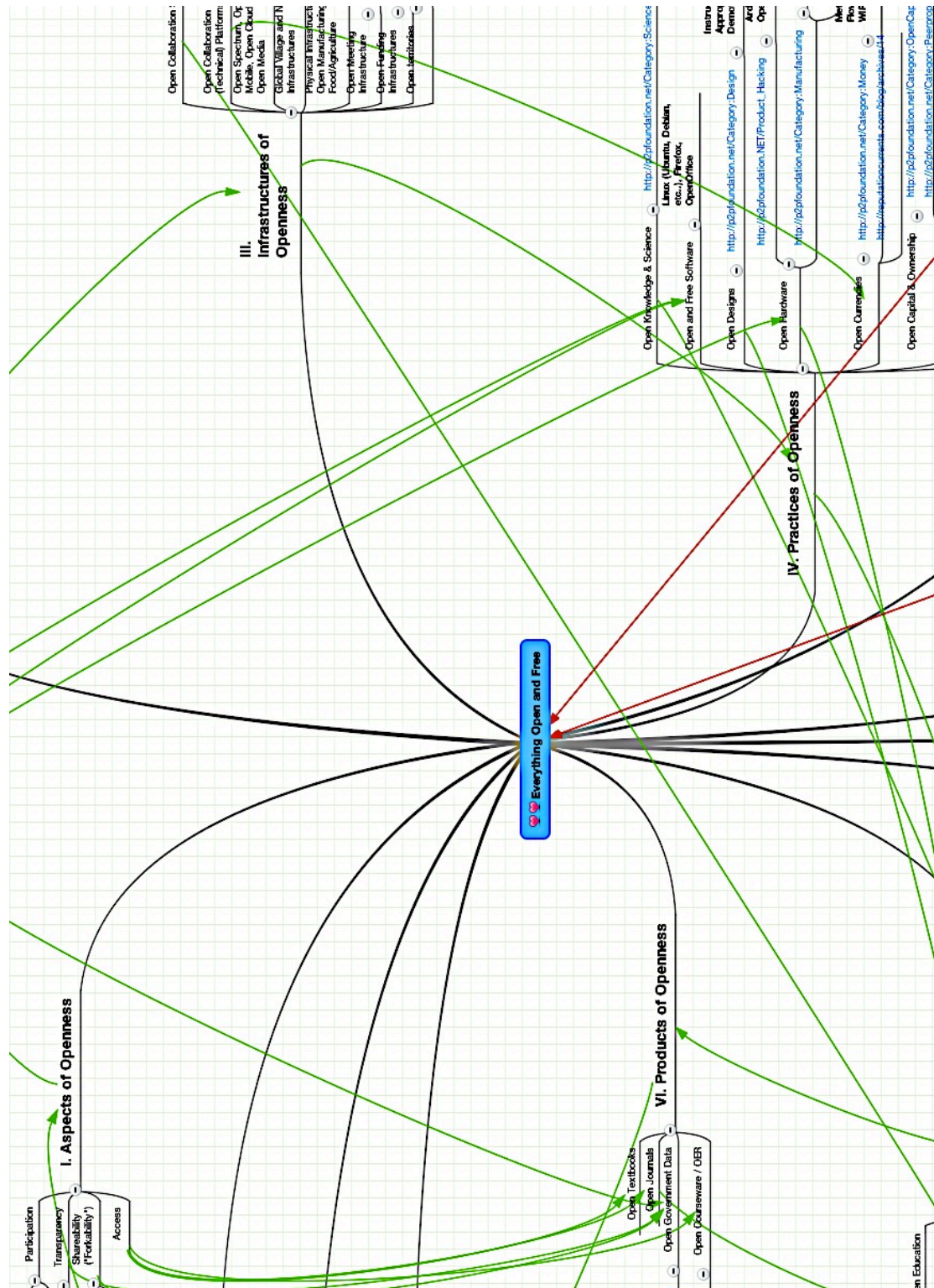


Figure 1

For Rosenberg, openness is also a quality of a system. Through competition, the most superior knowledge within that system will rise to the top and continue the march of the system's progress – 'better, more innovative products'.

I want to finish my review of contemporary openness by considering its deployment outside software, outside networks cultures, and into the realm of institutional politics and related writings. One of the first to translate the open (back) into institutional politics was Douglas Rushkoff. In 2003, he wrote a short monograph titled *Open source democracy: How online communication is changing offline politics* (Rushkoff, 2003). Rushkoff argues that a new 'electronic renaissance' has taken place, a profound shift in individuals' perceptions of their own agency in electronic environments. He uses open source as his key example: 'like literacy, the open source ethos and process are hard if not impossible to control once they are unleashed. Once people are invited to participate in, say, the coding of a software program, they begin to question just how much of the rest of the world is open for discussion' (Rushkoff, 2003: 56-57). Rushkoff's renaissance, however, does not merely detail how politics can benefit from the insights of open source; it is a politics totally enmeshed in computational metaphors: 'The implementation of an open source democracy will require us to dig deep into the very code of our legislative processes, and then rebirth it in the new context of our networked reality' (Rushkoff, 2003: 56).

Less than a decade later, Rushkoff's ideas are fast becoming the norm. For example, the recent edited collection, *Open government: Collaboration, transparency, and participation in practice* (Lathrop and Ruma, 2010), includes contributions by key members of government and commerce and is clearly aimed at a broad audience. The collection's rhetoric is a perfect mash up of Hayek, Popper, Raymond and Stallman, evidenced by a scan of the section titles such as, "Competition is critical to any ecosystem", "Open standards spark innovation and growth", "The closed model of decision making" and "Open government and open society". The connection to the organisational method of software cultures is made more explicit in Tim O'Reilly's contribution, "Government as a platform", where he writes:

What if... we thought of government as the manager of a marketplace? In *The cathedral & the bazaar*, Eric Raymond uses the image of a bazaar to contrast the collaborative development model of open source software with traditional software development, but the analogy is equally applicable to government. ... A bazaar... is a place where the community itself exchanges goods and services. (2010: 11)

The rest of the piece is dedicated to translating Raymond's insights to the practice of government. In another chapter, Charles Armstrong describes the profound impact the internet holds for democracy:

The Internet has changed a fundamental aspect of democratic systems which has persisted for 7,000 years. The change may presage a period of democratic innovation on a scale comparable to classical Greece. It will lead to democratic systems that are more fluid, less centralized, and more responsive than those we know today; systems where people can participate as little or as much as they wish and

where representation is based on personal trust networks rather than abstract party affiliations. This is Emergent Democracy. (2010: 167)

Upon considering ‘the road to emergent democracy’, Armstrong notes that ‘we tend to associate democracy with nations, cities, and other state entities’, but it is his ‘hunch that virtual corporations’ and not traditional institutions will pave the way (Armstrong, 2010: 175). The example of what these virtual institutions look like, once again, is Wikipedia. In both of these texts, government is re-imagined as a competitive marketplace of ideas, modelled after bazaar-like virtual corporations that resemble Wikipedia and which promise to reinvigorate democracy on a scale unmatched since classical Greece.

This kind of sentiment is similarly expressed by serving politicians and newly-established government initiatives. One early supporter of the open in Britain was the conservative MP, Douglas Carswell. In 2009, Carswell appeared on the current affairs show *Newsnight*, specifically in relation to the parliamentary expenses scandal, which later led to the stepping-down of the Speaker of the House of Commons. After commenting critically on the scandal, Carswell called for a move to “Open Source Politics”. This historic gesture was mirrored on his blog, where he writes: ‘Open source software. Wikipedia and wiki-learning. Open source parties and politics, too?’ (2009). He also writes about opening the primaries (the selection of candidates for election) – ‘Open primaries might spell the end for closed-shop parties’ (2009) – and name drops Clay Shirky and notions like ‘collaborative creation’.

Australia’s recent Government 2.0 Taskforce represents a more organised statement of political openness. On the “about” section of its website the Taskforce divides its stated areas of work into two streams:

The first relates to increasing the openness of government through making public sector information more widely available to promote transparency, innovation and value adding to government information.

The second stream is concerned with encouraging online engagement with the aim of drawing in the information, knowledge, perspectives, resources and even, where possible, the active collaboration of anyone wishing to contribute to public life. (2010)

Likewise, in a memorandum titled “Transparency and open government”, which introduced his Open Government Initiative in the United States, Barack Obama writes:

My Administration is committed to creating an unprecedented level of openness in Government. We will work together to ensure the public trust and establish a system of transparency, public participation, and collaboration. Openness will strengthen our democracy and promote efficiency and effectiveness in Government. (Obama, 2009)

Finally, even the recent conservative libertarian movement, the Tea Party, has borrowed ideals from open source. The production of the movement’s notorious “Contract from America”, a document that lists ten key agenda items the movement would like congress candidates to sign, was described by its creator in *The New York Times* as follows:

‘Hundreds of thousands of people voted for their favorite principles online to create the Contract as an open-sourced platform for the Tea Party movement’ (Becker, 2010). Openness, information, collaboration, transparency and participation – government 2.0 indeed.

Thus far, I have traced the open through debates about computer systems and standards, into software cultures in the eighties and nineties, and network cultures in the last decade, and, finally, outside of these realms and into that of institutional politics and society as a whole. And while the notion of openness in government is not unique to the current neo-liberal regime (as the writings of Popper make clear), I hope to have made clear that the general deployment of the open in institutional politics, and as a political concept more generally, cannot be separated from its emergence in software and network cultures. Indeed, it is perhaps more accurate to posit today’s openness as evidence of the networked and computational, even cybernetic, nature of governance. Through these multiple trajectories, openness is placed in a variety of settings; articulated alongside different concepts and put to use in different ways. The open circulates, scales up, garners new allies, is reconfigured, distinguished and remixed; each movement troubles and destabilises the articulation of its meaning. The open sways between means and ends, between noun, verb and adjective. And throughout all this movement, openness nonetheless maintains certain consistencies, such as its couplings with transparency, collaboration, competition and participation, and its close ties with various enactments of liberalism. What to make of a concept championed by all walks of political life? When conservative liberals, libertarians, liberal democrats, post-autonomous Marxists and left leaning activists all claim the open as their own and all agree that the open is the way forward? What to make of a politics that contains bits and pieces of many older political positions, but cannot be aligned easily with any one in particular? Or one that can be effortlessly deployed on any scale, from small projects to all society, by the American government or the radical P2P Foundation? One that defends markets but (at least in some instances) attacks (intellectual) property, and whose meaning is so overwhelmingly positive it seems impossible to question, let alone critique?

While the *force of the open* must be acknowledged – the real energy of the people who rally behind it, the way innumerable processes have been transformed in its name, the new projects and positive outcomes it has produced – I suggest that the concept itself has some crucial problems. In the final section of this essay, I aim to demonstrate that not only is the open problematic in relation to contemporary software and network cultures, but that the concept contains a poverty that has existed in all its uses throughout history and that makes it unsuitable for political description. Indeed, I argue that the open actively works against the development of a political language – if, that is, we take the political to extend beyond questions of just governance to the circulation and distribution of power and force, and take politics to mean the distributions of agency in general, as well as the conflicts and issues that emerge when antagonistic flows intersect. To make this argument, I return to Popper and *The open society*.

A critique of the open

The various criticisms Popper levels against Plato and his followers have not gone unchallenged (Kendall, 1960; Cornforth, 1968; Levinson, 1970; Vernon, 1976; Shearmur, 1996; De Cock and Bohm, 2007). The critique I put forward does not rely on whether Popper was faithful in his interpretation of the great philosophers, or of the terrible events that inspired his texts, as do others. I do not interrogate how Popper's political philosophy is intimately connected to, if not derived from, his scientific philosophy and I do not challenge the validity of the unique concepts he relies on in *The open society*, such as critical rationalism and piecemeal social engineering. Instead, I focus specifically on the character of openness *per se* in relation to the rest of the text. I argue that the logic of openness actually gives rise to, and is perfectly compatible with, new forms of closure; indeed, that closure is inherent in Popper's notion of openness. Moreover, I claim that there is something about openness, about the mobilisation of the open and its conceptual allies, that actively works against making these closures visible. I finish by reflecting on the peculiar situation of the second coming of openness within the supposedly already-open society.

Like his critique of the closed – which has both philosophical and empirical dimensions – Popper's open society is both a set of ideas and a really existing entity. These two dimensions, which enact an awkward distinction between reality and thought, play out and interact in complex and ultimately troubled ways. The first thing one notices about *The open society* is that it is almost entirely dedicated to critique. In the first volume, for example, the open is not taken up at all until the final chapter, after nine preceding chapters on Plato. It is a work explicitly concerned with the *Enemies* of the open, rather than the open itself (Magee, 1982: 87). As a concept, therefore, the open is reactionary; it gains meaning largely through a consideration of what it is *not*. The open is significant in terms of the actually existing political situation of Popper's time largely because it is neither fascism nor communism. As we shall see, this negative, or *is not*, quality of the concept and concomitant reluctance to build a lasting affirmative dimension is entirely necessary and gets to the heart of the problem of the open as a political concept.

Of the negative or *is not* qualities of the open, we can extract the following from Popper's critique of closed societies: open societies do not condone historical or economic determinism; do not support programs of radical social engineering based on truth claims; and do not hold any truth to be absolute. These qualities are the direct result of the critique of the closed I detailed earlier. Under closer analysis, however, it is possible to identify numerous positive qualities of open societies, even though these are generally mentioned in passing and without extended elaboration in the text. Such positive qualities surface especially in the final chapter of the first volume, after the critique of Plato is concluded. In these pages Popper regularly invokes reason and the rational as characteristics of open societies, noting for example that the open is 'a rational attempt to improve social conditions' (1962: 172). Open societies are also individual-centric in terms of decision-making, responsibility, competition and familial ties. According to Popper, 'Personal

relationships of a new kind can arise where they can be freely entered into, instead of being determined by the accidents of birth; and with this, a new individualism arises' (1962: 175); and 'the society in which individuals are confronted with personal decisions, [is] the *open society*' (1962: 173). Such individualism also leads to strong competition and exchange relations: 'one of the most important characteristics of the open society [is] competition for status among its members,' and further, 'in an open society, many members strive to rise socially, and to take the place of other members. This may lead, for example, to such important social phenomenon as class struggle' (1962: 174). Regarding exchange, Popper writes that 'our modern open societies function largely by way of abstract relations, such as exchange or co-operation' (1962). While discussing the historical beginnings of the open society, Popper offers the following quasi-legal and ethical characteristics: 'The new faith of the open society, the faith in man [sic], in equalitarian justice, and in human reason, was perhaps beginning to take shape, but it was not yet formulated' (1962: 189). Furthermore, he states, 'Individualism, equalitarianism, faith in reason and love of freedom were new, powerful, and, from the point of view of the enemies of the open society, dangerous sentiments that had to be fought' (1962: 199). Finally, throughout the text as a whole, Popper regularly gestures towards a preference for democracy over totalitarian regimes. Indeed, at times the open society appears interchangeable with Popper's understanding of democracy.

Of these negative and positive identifiers, it is only the negative qualities that approach anything like the essence or definitive core of openness. More precisely, the positive qualities of openness are actually negative qualities masked as positive ones, or alternatively exist at the level of reality (of real practices) and are therefore subject to continual transformation. Openness emerges as a theory bereft of content coupled with a really existing practice, defined by its continual non-identification with itself. This character of openness is made clear through a consideration of Popper's understanding of democracy – one of the key *positive* qualities of the open society. He writes:

The theory I have in mind is one which does not proceed, as it were, from a doctrine of the intrinsic goodness or righteousness of a majority rule, but rather from the baseness of tyranny; or more precisely, it rests upon the decision, or upon the adoption of the proposal, to avoid and resist tyranny. (Popper, 1962: 124)

Popper proceeds to immediately distinguish between two 'types of government': one that can be removed 'without bloodshed', through institutional processes such as elections; the other which can only be removed via revolution. Only the first mode of governance is democratic: 'I suggest the term 'democracy' as a short-hand label for a government of the first type, and the term 'tyranny' or 'dictatorship' for the second'. So, a 'theory of democracy' on the one hand, a theory defined only as against tyranny, or *not tyranny*, and a practice of democracy on the other, with institutions and processes and all the messy details that practice implies. This *practice* of democratic governance, though, and as described, is necessarily dynamic and only ever represents one more or less flawed instantiation, one better or worse implementation of the theory of "*not tyranny*":

[T]he various equalitarian methods of democratic control, such as general elections and representative government, are to be considered as no more than well-tried and, in the presence of a widespread traditional distrust of tyranny, reasonably effective institutional safeguards against tyranny, always open to improvement, and even providing methods for their own improvement. (Popper, 1962: 125)

Democracy, as we can see, is only a positive quality of openness to the extent that it replicates the form of openness itself, in a fractal manner: it is composed of a negative core and an ephemeral, positive reality. Democracy's existence as a positive quality of the open is therefore at best dubious. As I have suggested, these two related components of openness, each of which defers the concept's meaning in different ways, are in fact necessary. Indeed, it is the continual deferral, the reluctance to assert an authoritative positive identity, which distinguishes the open from totalitarian, closed forms of knowledge. The open is a kind of negative response to totalitarian knowledge; a recognition of the fundamental agential significance of knowledge, of the old power/knowledge coupling, and at the same time an attempt to insert some distance between them by emptying one side of the equation.

I want to suggest at this point that all of these positive, if ephemeral, traits of the open are alive and well in contemporary society; they remain, in fact, the central values of neo-liberal democracies: freedom, democracy, individualism, competition and exchange (free markets), equalitarian justice and reason. This is hardly surprising, considering Popper's (abstract) concept of the open society in fact corresponds with the actually existing capitalist democracies of the mid-20th century, and such societies persist. (This is not to suggest that such characteristics are indisputably realised or haven't changed their meaning and function, but rather that politics still plays out in and through this language). It seems, therefore, that we are still in the society Popper described as open, both in terms of *is not communism or fascism*, and in terms of the positive qualities just mentioned. At the same time, however, and consistent with Popper's logic, this actually existing open society is likely to change.

To continue with the account of democratic practices, for example, it is likely, and indeed categorically necessary, that these practices may be replaced. How is it that *specific sets of practices* called 'democracy' are part of the open and yet in future might not be? One possible response is that the democratic practices might be succeeded by something that is even more democratic, and thus, even more open. Another possible response is that these practices have *become* closed; that somehow, through time this mode of governance loses its character of openness. Both of these responses suggest that *forms of closure* exist *within* open societies: in the first scenario, if a practice is to be replaced with a *more open* one, it must not be entirely open to begin with. The second scenario indicates that the seeds of closure are already immanent within this open mode of governance. However, the alternative, which is to affirm a specific version of democracy and a specific programme of knowledge and related practices – in short, a precise truth of the open – is simultaneously the open's closure. Thus, by invoking positive but ephemeral qualities, and a society that necessarily changes, Popper avoids the kinds of closure he identifies in totalitarian thought.

At the same time, closure remains an inherent part of the open; it is what openness must continually respond to and work against – a continual threat amongst the ranks.

Openness, we might say, implies antagonism, or what the language of openness would describe as closures. Such closures do not randomly emerge, unexpected and from the outside. It is the very qualities that Popper holds up as representative of contemporary openness and which constitute the formal language of the just organisation of society – freedom, competition, equality and exchange coupled with democratic institutions – that not only coincide with, but are actually generative of new forms of closure. The most obvious of these are the ‘economic’ closures produced by ‘competition’ and ‘free markets’, continuous and generalised asymmetrical distributions of agency produced by (debt-based) informational capitalism of which the on-going global financial crisis is only the most recent and dramatic episode. These are the same conditions that produced the invisible source code, non-disclosure agreements and broader regime of intellectual property that Stallman experienced as closures in his lab at MIT. In short, Popper’s argument against totalitarian knowledge – replicated faithfully by his close friend and intellectual ally Hayek to defend free markets and private property over centralised planning – is compatible with and even constitutive of neo-liberal capitalism. And it is these same forms of closure that the second coming of openness, together with its new set of conceptual allies, tries to address. But what to make of this second coming?

The first thing one notices is the curious situation of openness emerging within a supposedly already-open society. Other than confirming the closures inherent in the open, I think this curious situation is suggestive of a crucial conceptual shortcoming of openness. Once an organisation, state or project is labelled open, it becomes difficult to *account for* the politics (closures) that emerge from within. For Popper, this is because his version of the open is primarily a critique of totalitarian knowledge, but also because he struggles to focus on the details of his open society for fear of closing it. Recent uses of openness – from open systems, to open source and free and open everything – bear significant resemblances to Popper’s in terms of character and function. Once more, the open emerges largely as a reaction to a set of undesirable developments, beginning with the realm of closed systems and intellectual property and its ‘closed source’. And once more, the open is articulated alongside an entourage of fractal sub-concepts that defer political description: participation, collaboration and transparency. While this re-emergence works as a critique of Popper-Hayek openness, it simultaneously reinstates the same conceptual architecture. Of all the authors cited in the account of openness I have developed here, for example, not one has turned a critical eye upon the open, and there has been very little criticism of specific open projects. If a critical word is written, it is never about the whole, but only about how one small component can be made better, more open. Somewhat ironically, once something is labeled open, it seems that no more description is needed. Openness is the answer to everything and it is what we all agree upon.

I began this essay by quoting Lessig and Hardt and Negri, radically different thinkers who both gesture towards the open. Throughout my analysis I added many names and

organisations to these three, from the Tea Party to leftist activist groups, governments, major corporations and scholars. All these individuals and groups understand their practices and ideas in relation to the open and use it to 'look forward'. I hope to have shown, however, that the open has not proven well suited to this task. Rather than using the open to look forward, there is a need to look more closely at the specific projects that operate under its name – at their details, emergent relations, consistencies, modes of organising and stabilising, points of difference, and forms of exclusion and inclusion. If we wish to understand the divergent political realities of things described as open, and to make visible their distributions of agency and organising forces, we cannot 'go native', as a young, anthropologically-minded Bruno Latour once wrote, meaning that we cannot adopt the language used in the practices we wish to study. To describe the political organisation of all things open requires leaving the rhetoric of open behind.

references

- Armstrong, C. (2010) 'Emergent democracy', in D. Lathrop and L. Ruma (eds.) *Open government: Collaboration, transparency, and participation in practice*. Cambridge: O'Reilly Media.
- Becker, B. (2010) 'A revised contract for America, minus "with" and Newt', *The New York Times*, 15 April.
- Bentham Open (2010) 'Welcome to Bentham open access' [<http://www.bentham.org/open/index.htm>].
- Bergson, H., R. A. Audra, C. Brereton and W. H. Carter (1935) *The two sources of morality and religion*. New York: H. Holt and Company.
- Berry, D. M. (2008) *Copy, rip, burn: The politics of copyleft and open source*. London: Pluto.
- Bertalanffy, L. V. (1960) *Problems of life: An evaluation of modern biological and scientific thought*. New York: Harper & Brothers.
- Bezroukov, N. (1999) 'Open source software development as a special type of academic research (critique of vulgar Raymondism)', *First Monday*, 4(10).
- Carswell, D. (2009) 'Open the party; open source politics is coming', *Douglas' Blog* [<http://www.talkcarswell.com/show.aspx?id=926>].
- Coleman, G. (2004) 'The political agnosticism of free and open source software and the inadvertent politics of contrast', *Anthropology Quarterly*, 77(03): 507-519.
- Cornforth, M. (1968) *The open philosophy and the open society; A reply to Dr. Karl Popper's refutations of Marxism*. New York: International Publishers.
- De Cock, C. and Bohm, S. (2007) 'Liberalist fantasies: Zizek and the impossibility of the open society', *Organization*, 14(6): 815-836.
- Free Software Foundation. (2010) 'When a company asks for your copyright' [<https://www.gnu.org/philosophy/assigning-copyright.html>].
- Ghosh, R. A. and V. V. Prakash (2000) 'The Orbitan free software survey', *First Monday*, 5(7).
- Government 2.0 Taskforce. (2010) 'About' [<http://gov2.net.au/about/>].
- Hardt, M. and A. Negri (2004) *Multitude: War and democracy in the age of empire*. New York: The Penguin Press.
- Hayek, F. A. V. (1944) *The road to serfdom*. London: Routledge.
- Kelty, C. M. (2008) *Two bits: The cultural significance of free software*. Durham: Duke University Press.
- Kendall, W. (1960) 'The "open society" and its fallacies', *The American Political Science Review*, 54(4): 972-979.
- King, J. (2006) 'Openness and its discontents', in J. Dean, J. W. Anderson and G. Lovink (eds.) *Reformatting politics: Information technology and global civil society*. New York: Routledge.

- Lathrop, D. and L. Ruma (eds.) (2010) *Open government: Collaboration, transparency, and participation in practice*. Cambridge: O'Reilly Media.
- Lessig, L. (2005) 'Open code and open societies', in J. Feller, B. Fitzgerald, S. A. Hissam and K. R. Lakhani (eds.) *Perspectives on free and open source software*. London: The MIT Press.
- Lessig, L. (2008) *Remix: Making art and commerce thrive in the hybrid economy*. New York: Penguin Press.
- Levinson, R. B. (1970) *In defense of Plato*. New York: Russell & Russell.
- Magee, B. (1982) *Popper*. London: Fontana.
- MIT Open Courseware. (2010) 'MIT Open Courseware' [<http://ocw.mit.edu/>].
- Obama, B. (2009) 'Transparency and open government' [http://www.whitehouse.gov/the_press_office/TransparencyandOpenGovernment/].
- O'Reilly, T. (2010) 'Government as a platform', in D. Lathrop and L. Ruma (eds.) *Open government: Collaboration, transparency, and participation in practice*. Cambridge: O'Reilly Media.
- Open Courseware Consortium (2010) 'Open courseware consortium' [<http://www.ocwconsortium.org>].
- Open Everything Contributors (2010) 'Open everything' [<http://openeverything.wikis.org>].
- Open Humanities Press (2010) 'About: Principles and goals' [<http://openhumanitiespress.org/goals.html>].
- Open Humanities Press (2010) 'Open humanities press' [<http://openhumanitiespress.org/index.html>].
- Open Medicine (2010) 'Mission' [<http://www.openmedicine.ca/pages/view/Mission>].
- Open Medicine (2010) 'Open access' [<http://www.openmedicine.ca/pages/view/Open%20Access>].
- P2P Foundation Contributors (2010) 'The foundation for P2P alternatives' [http://p2pfoundation.net/The_Foundation_for_P2P_Alternatives].
- Plato (1974) *The Republic*. Harmondsworth; Baltimore: Penguin.
- Popper, K. R. (1962) *The open society and its enemies*. London: Routledge and Kegan Paul.
- Raymond, E. S. (2000) 'The cathedral and the bazaar' [<http://catb.org/~esr/writings/homesteading/>].
- Rosenberg, J. (2009) 'The meaning of open' [<http://googlepublicpolicy.blogspot.com/2009/12/meaning-of-open.html>].
- Rushkoff, D. (2003) *Open source democracy: How online communication is changing offline politics*. London: Demos [<http://rushkoff.com/books/open-source-democracy>].
- Shearmur, J. (1996) *The political thought of Karl Popper*. London; New York: Routledge.
- The Open Knowledge Foundation (2010) 'Projects' [<http://www.okfn.org/projects>].
- The Open Knowledge Foundation (2010) 'About Us' [<http://www.okfn.org/about>].
- The Open Knowledge Foundation (2010) 'Our Vision' [<http://www.okfn.org/vision>].
- Tkacz, N. (2010) 'Wikipedia and the politics of mass collaboration', *Platform: Journal of Media and Communication*, 2(2).
- Vernon, R. (1976) 'The "great society" and the "open society": Liberalism in Hayek and Popper', *Canadian Journal of Political Science*, 9(2): 261-276.
- Wikipedia Contributors (2010) 'Wikipedia: About', *Wikipedia: The Free Encyclopaedia* [<http://en.wikipedia.org/wiki/Wikipedia:About>].
- Williams, S. (2002) *Free as in freedom: Richard Stallman's crusade for free software*. Sebastopol, CA: O'Reilly & Associates, Inc.

the author

Nathaniel Tkacz is an Assistant Professor at The University of Warwick. He is co-editor (with Geert Lovink) of *Critical Point of View: A Wikipedia Reader* (2011). Details of his research can be found at nathanieltkacz.net.
E-mail: n.tkacz@warwick.ac.uk